MA388 Sabermetrics: Lesson 19

Comparing Peak Trajectories - Part II

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Comparing Trajectories of Home Run Hitters

1. Find the ten players in baseball history who have hit the most career home runs.

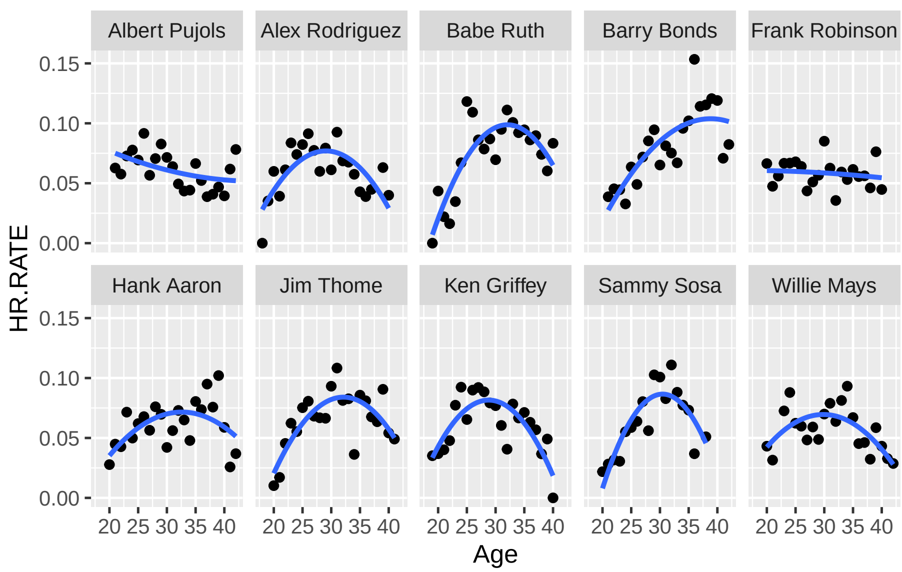
```
library(tidyverse)
library(Lahman)
library(knitr)
```

```
Top10 <- Batting |>
  summarize(totalHR = sum(HR), .by ='playerID') |>
  slice_max(totalHR,n=10) |>
  inner_join(People, by = "playerID") |>
  mutate(label = str_c(nameFirst,nameLast,sep=' ')) |>
  relocate(label,totalHR)

Top10 |>
  select(label, totalHR, debut) |>
  kable()
```

label	totalHR	debut
Barry Bonds	762	1986-05-30
Hank Aaron	755	1954-04-13
Babe Ruth	714	1914-07-11
Albert Pujols	703	2001-04-02
Alex Rodriguez	696	1994-07-08
Willie Mays	660	1951-05-25
Ken Griffey	630	1989-04-03
Jim Thome	612	1991-09-04
Sammy Sosa	609	1989-06-16
Frank Robinson	586	1956-04-17

2. Fit quadratic functions to the home run rates of the ten players, where HR.RATE = HR / AB. Display the fitted trajectories in a single figure.

```
top10_stats <- Batting |>
  filter(playerID %in% Top10$playerID) |>
  summarize(HR.RATE = sum(HR)/sum(AB),.by=c('playerID','yearID')) |>
  inner_join(People, by = "playerID") |>
  mutate(birthyear = ifelse(birthMonth >= 7,
                            birthYear + 1, birthYear),
         label = str_c(nameFirst,nameLast,sep=' '),
         Age = yearID - birthyear)

top10_stats |>
  ggplot(aes(x = Age, y = HR.RATE)) +
  geom_point() +
  geom_smooth(method = "lm", formula = y~poly(x,2), se = FALSE) +
  facet_wrap(~label, ncol = 5)
```



3. On the basis of your work, which player had the highest estimated home run rate at his peak? Which player among the ten had the smallest peak home run rate?

```
top10_model <- function(this_playerID){

  this_player_stats <- top10_stats |>
```

```r
    filter(playerID == this_playerID)

  fit <- lm(HR.RATE ~ I(Age - 30) + I((Age - 30)^2), data = this_player_stats)

  A <- coef(fit)[1]
  B <- coef(fit)[2]
  C <- coef(fit)[3]

  # Return all the stuff you just obtained.
  tibble(label = head(this_player_stats$label,1), playerID = this_playerID, A, B, C,
         age.max = 30 - B / C / 2,
         max = A - B^2 / C / 4)
}

# Map the model function to the list of playerIDs you obtained in 1.
HR_results <- map_df(Top10$playerID,top10_model)
HR_results |>
  kable()
```

label	playerID	A	B	C	age.max	max
Barry Bonds	bondsba01	0.0851324	0.0042375	-0.0002401	38.82412	0.1038285
Hank Aaron	aaronha01	0.0700492	0.0011897	-0.0002294	32.59352	0.0715920
Babe Ruth	ruthba01	0.0965013	0.0022230	-0.0005377	32.06727	0.0987990
Albert Pujols	pujolal01	0.0610317	-0.0012091	0.0000387	45.60271	0.0515987
Alex Rodriguez	rodrial01	0.0766162	-0.0007482	-0.0003996	29.06393	0.0769664
Willie Mays	mayswi01	0.0695671	-0.0000923	-0.0002813	29.83585	0.0695747
Ken Griffey	griffke02	0.0808547	-0.0012131	-0.0005022	28.79225	0.0815873
Jim Thome	thomeji01	0.0820880	0.0018386	-0.0004322	32.12711	0.0840435
Sammy Sosa	sosasa01	0.0864312	0.0006562	-0.0007198	30.45580	0.0865808
Frank Robinson	robinfr02	0.0586096	-0.0002988	-0.0000104	15.61690	0.0607582

```r
# Highest estimated max HR rate:
HR_results |>
  slice_max(max)
```

```
# A tibble: 1 x 7
  label       playerID       A       B         C age.max   max
  <chr>       <chr>      <dbl>   <dbl>     <dbl>   <dbl> <dbl>
1 Barry Bonds bondsba01 0.0851 0.00424 -0.000240    38.8 0.104
```

```r
# Lowest estimated max HR rate:
HR_results |>
  slice_min(max)
```

```
# A tibble: 1 x 7
  label         playerID       A        B         C age.max    max
  <chr>         <chr>      <dbl>    <dbl>     <dbl>   <dbl>  <dbl>
1 Albert Pujols pujolal01 0.0610 -0.00121 0.0000387    45.6 0.0516
```

Your answer here.

4. Do any of the players have unusual career trajectory shapes? Is there any possible explanation for these unusual shapes?

Your answer here.